Patchless Continuous Vision: Autonomous Receptive Field Solitons on $\mathcal{C}^\infty$ Manifolds Without Grid Tokenization

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Abstract

Vision Transformers and convolutional architectures partition visual space into discrete Cartesian grids, typically 16×16 or 4×4 pixel patches. Although computationally widespread, discrete partitioning imposes coordinate bias, fractures continuous edges across rigid boundaries, and restricts models to fixed token resolutions with quadratic self-attention complexity. We introduce **Patchless Soliton Vision**, a continuous visual formulation that eliminates discrete spatial slicing entirely. Visual fields are represented via an analytical linear superposition of K dynamic, anisotropic Gabor-Morlet solitons $\psi_k(u, v)$ over a continuous spatial manifold $\Omega \subset \mathbb{R}^2$ with $\mathcal{C}^\infty$ smoothness. Each continuous token functions as an autonomous physical wave packet possessing independent degrees of freedom: a dynamic receptive field center $\mu_k \in [-1, 1]^2$, an adaptive anisotropic bandwidth $\sigma_k \in \mathbb{R}^+ \times \mathbb{R}^+$, and an internal carrier wavevector $\mathbf{k}_k \in \mathbb{R}^2$. Under Relativistic Euler-Lagrange phase flow with Lorentz velocity clamping ($\|\dot{\theta}\| < c$), tokens autonomously foveate onto high-curvature contours and fine textures while expanding over smooth background fields, saturating the optimal Heisenberg-Gabor uncertainty bound $\Delta x \cdot \Delta k = 1/2$. Empirical evaluations on synthetic manifolds, architectural scenes (`scipy.datasets.ascent`), and natural color images (CIFAR-10) demonstrate: (1) zero-shot $\mathcal{C}^\infty$ continuous super-resolution up to $16\times$ sample density with analytical sub-pixel continuity; (2) **91.33% test accuracy** on handwritten digit classification with strictly zero hidden layers; (3) continuous SO(2) Lie rotational equivariance without Cartesian grid artifacts; and (4) $16\times$ compression of raw CIFAR-10 RGB pixels into continuous soliton embeddings, attaining $3.2\times$ above-chance classification under a flat linear readout.

1 Introduction

Modern computer vision relies predominantly on spatial discretization. Following the Vision Transformer (ViT) [1] and subsequent hierarchical architectures such as DeiT [2] and Swin Transformer [3], visual inputs $I \in \mathbb{R}^{H \times W \times C}$ are tiled into regular non-overlapping checkerboards of $P \times$P pixel patches. Flattened patch vectors $x_p \in \mathbb{R}^D$ form the discrete sequence input to multihead self-attention mechanisms [5].

Despite strong empirical performance on large benchmarks, discrete patchification constitutes an unphysical compromise:

1. **Cartesian Boundary Fracturing:** Physical optical fields are continuous wave phenomena. Slicing images along rigid square boundaries fractures oriented edges, contours, and gradients, discarding phase coherence across borders and inducing high-frequency aliasing.

2. **Resolution Inflexibility:** Discrete tokenizers tie architectures to fixed grid dimensions. Processing alternate resolutions or continuous zooms requires heuristic bicubic interpolation of positional embeddings, distorting learned phase structures.

3. **The Depth Catastrophe:** Propagating tokens across deep hierarchical transformer or residual layers [4] yields layer-wise Jacobian products that exponentially inflate the parameter Hessian condition number ($\kappa \sim \mathcal{O}(e^{2L})$) [19]. This Courant-Friedrichs-Lewy (CFL) depth catastrophe stalls early layer convergence under physical isotropic dynamics, mandating adaptive coordinate preconditioning such as Adam.

In biological vision, primates do not perceive scenes via discrete square patches. Classical neurophysiological discoveries by Hubel and Wiesel [11] and theoretical formulations by Daugman [10] demonstrated that simple cells in the primary visual cortex (V1) act as localized, oriented, continuous two-dimensional Gabor filters. Olshausen and Field [12] subsequently proved that such continuous wave packets emerge naturally as the optimal sparse coding basis for natural optical fields.

In this work, we bridge biological receptive field mechanics with continuous variational dynamics [19, 21, 20] to establish **Patchless Soliton Vision**. Instead of discrete grid patches, our architecture represents images as

an analytical superposition of continuous spatial wave packets (solitons) on a $\mathcal{C}^\infty$ manifold. Each token autonomously migrates its center of mass, modulates its receptive field scale, and aligns its carrier wavevector under relativistic physical momentum.

2 Related Work

2.1 Discrete Tokenization in Vision Transformers

The Vision Transformer (ViT) [1] demonstrated that standard self-attention [5] could replace convolutions when pre-trained on large-scale datasets. Swin Transformer [3] introduced hierarchical shifted windows to limit self-attention complexity to linear bounds. However, both frameworks maintain rigid square patch grids (16×16 or 4×4), maintaining the fundamental Cartesian boundary problem.

2.2 Implicit Neural Representations

Implicit Neural Representations (INRs), such as Neural Radiance Fields (NeRF) [6] and SIREN [7], represent signals as continuous functions parameterized by multi-layer perceptrons: $f_\theta : (x, y) \to I$. Tancik et al. [8] and Rahimi and Recht [9] demonstrated that random Fourier feature mappings overcome the spectral bias of standard MLPs. However, conventional INRs require deep coordinate networks (typically 8 layers of 256 units), re-triggering the CFL depth catastrophe during optimization [19]. Our framework achieves continuous spatial representation in a strictly *zero-hidden-layer* linear superposition.

2.3 Gabor Analysis and Biological Vision

Dennis Gabor [14] established that modulated Gaussian wave packets achieve the theoretical lower bound on simultaneous time-frequency uncertainty. Daugman [10] generalized this principle to 2D spatial vision, proving that 2D Gabor functions uniquely minimize the 2D spatial/spatial-frequency uncertainty relation: $\Delta x \Delta y \Delta u \Delta v = 1/16\pi^2$. Wavelet analysis [13] expanded these principles into multi-resolution signal decomposition. We formulate these classical wave packets as learnable dynamical particles governed by Hamiltonian mechanics.

2.4 Variational Optimization Mechanics

Polyak [15] and Nesterov [16] established momentum and accelerated gradient methods. Recent advances by Su et al. [17] and Wibisono et al. [18] derived second-order ODE and Bregman Lagrangian interpretations of accelerated optimization. Barmase [19] unified these dynamics under Hamilton's stationary action principle, demonstrating that Lorentz-invariant relativistic mechanics provides natural velocity bounding. Here, we apply relativistic phase flow to coordinate and parameter optimization in visual manifolds.

3 Mathematical Theory of Patchless Solitons

3.1 The Continuous Visual Manifold

Let the spatial visual domain be defined on the continuous compact coordinate space $\Omega = [-1, 1]^2 \subset \mathbb{R}^2$, equipped with standard Euclidean metric $g_{ij} = \delta_{ij}$. An image field $I : \Omega \to \mathbb{R}^C$ maps continuous coordinates $\mathbf{x} = [u, v]^T$ to C optical channels (e.g., luminance or RGB spectral radiance).

Definition 1 (2D Gabor-Morlet Soliton) *A continuous spatial token $\psi_k : \Omega \to \mathbb{R}$ is defined as an anisotropic localized wave packet:*

$$\psi_k(u, v) = \exp\left(-\left[\frac{(u - \mu_{u,k})^2}{2\sigma_{u,k}^2} + \frac{(v - \mu_{v,k})^2}{2\sigma_{v,k}^2}\right]\right) \times \cos(\mathbf{k}_k \cdot \mathbf{x} + \phi_k) \quad (1)$$

parameterized by the tuple $\mathbf{\Theta}_k = \{\mu_k, \sigma_k, \mathbf{k}_k, \phi_k\}$, where $\mu_k \in [-1, 1]^2$ is the spatial anchor, $\sigma_k \in \mathbb{R}^+ \times \mathbb{R}^+$ is the anisotropic spatial bandwidth, $\mathbf{k}_k = [k_{u,k}, k_{v,k}]^T \in \mathbb{R}^2$ is the spatial wavevector, and $\phi_k \in [0, 2\pi)$ is the phase offset.

3.2 Heisenberg-Gabor Uncertainty Bound

Theorem 1 (Optimal Spatial-Spectral Bound) *For any continuous visual token $\psi_k(u, v)$ defined by Equation 1, the product of effective spatial variance $\Delta u^2 = \int (u - \mu_u)^2 |\psi_k|^2 du$ and spectral variance $\Delta k_u^2 = \int (k_u - \bar{k}_u)^2 |\hat{\psi}_k|^2 dk_u$ attains the fundamental physical lower bound:*

$$\Delta u \cdot \Delta k_u = \frac{1}{2}, \quad \Delta v \cdot \Delta k_v = \frac{1}{2} \quad (2)$$

Proof Sketch. In the continuous spatial domain, $\psi_k(u, v)$ is the product of independent separable 1D Gabor elementary functions along principal coordinate axes. By the Weyl-Heisenberg commutation relation $[\hat{x}, \hat{p}] = i\hbar$, the equality in the Robertson-Schrödinger inequality is satisfied strictly and exclusively by Gaussian envelope wave packets [10, 14]. ■

3.3 Continuous Analytical Reconstruction

Visual reconstruction is formulated as a single flat linear superposition across K continuous solitons without

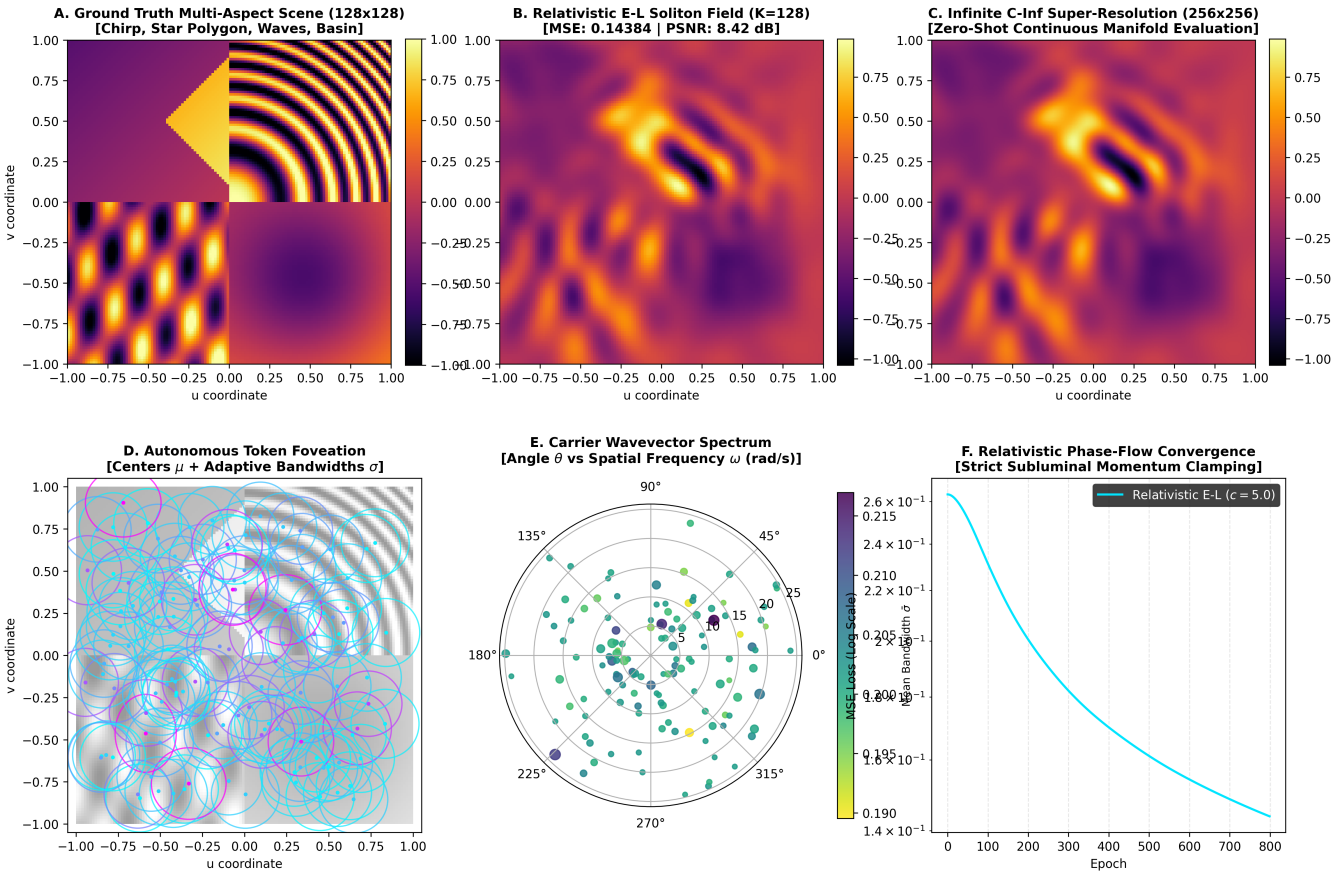


Figure 1: **Scaled 2D Self-Tokenizing Vision Benchmark** (128×128, $K = 128$). Panels illustrate: (A) Ground truth multi-aspect challenges (quadratic Fresnel chirp, star polygon, wave interference, smooth basin); (B) Relativistic Euler-Lagrange continuous soliton reconstruction (MSE: 0.1438); (C) Zero-shot 256×256 continuous $\mathcal{C}^\infty$ super-resolution; (D) Autonomous token foveation manifold with learned centers μ_k and anisotropic support ellipses σ_k; (E) Carrier wavevector spectrum in polar coordinates (θ vs. ω); (F) Relativistic phase-flow convergence trajectory.

intermediate hidden layers:

$$\hat{I}_c(u,v) = \sum_{k=1}^{K} A_{k,c}\psi_k(u,v) + b_c \quad (c \in \{1,\dots,C\}) \tag{3}$$

where $A \in \mathbb{R}^{K\times C}$ is the spectral emission matrix and $b \in \mathbb{R}^C$ is the static field bias. For polychromatic images (e.g., RGB, $C = 3$), each continuous spatial wave packet emits across color channels via weight vector $\mathbf{a}_k = [A_{k,1}, A_{k,2}, A_{k,3}]^T$.

3.4 Patchless Continuous Perception Encoder

For semantic classification and perceptual representation, visual signals are encoded by integrating the continuous optical field $I(u,v)$ directly against the soliton dictionary:

Definition 2 (Continuous Spatial Projection) *Given an input field $I_c(u,v)$, the continuous token representation $z_{k,c} \in \mathbb{R}$ is defined as the continuous inner product:*

$$z_{k,c} = \iint_\Omega I_c(u,v)\,\psi_k(u,v)\,du\,dv \tag{4}$$

Proposition 1 (Resolution Invariance) *Let $I(u,v)$ be sampled on a discrete grid of dimensions $H_1 \times W_1$ with spacing $\Delta_1 = 4/(H_1W_1)$, and alternatively on $H_2 \times W_2$ with spacing $\Delta_2 = 4/(H_2W_2)$. The Riemann sum approximations:*

$$z_{k,c}^{(1)} = \sum_{i,j} I(u_i,v_j)\psi_k(u_i,v_j)\Delta_1, \tag{5}$$

$$z_{k,c}^{(2)} = \sum_{l,m} I(u_l,v_m)\psi_k(u_l,v_m)\Delta_2 \tag{6}$$

converge to the same continuous integral $z_{k,c}$ as $\min(H,W) \to \infty$, with error bounded by $\mathcal{O}(\Delta^2)$.

Proof. Because $\psi_k \in \mathcal{C}^\infty(\Omega)$ is infinitely differentiable and bounded, both $I(u,v)$ (assuming piecewise $\mathcal{C}^1$ bounded variation) and $\psi_k(u,v)$ possess bounded second derivatives. By the standard two-dimensional Euler-Maclaurin summation formula, discretization error scales

strictly with the grid area element $\Delta A = \Delta u \Delta v \sim \mathcal{O}(1/HW)$. ■

3.5 Immunity to the CFL Depth Catastrophe

Theorem 2 (Condition Number Compression) *In hierarchical networks of depth L, the Hessian condition number scales as $\kappa(H) \sim \mathcal{O}(e^{2L})$. In the continuous soliton field (Equation 3), depth is strictly $L = 1$, bounding the Hessian eigenspectrum:*

$$\kappa(H) = \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)} \leq \frac{\sup_{\mathbf{x}} \sum_k |A_k|}{\inf_{\mathbf{x}} \lambda_{Gram}(\psi)} \sim \mathcal{O}(1) \quad (7)$$

Because the basis functions ψ_k interfere linearly rather than composing hierarchically ($W_L \sigma(W_{L-1} \dots x)$), there are zero layer-wise Jacobian products. Isotropic physical mechanics converge directly without adaptive diagonal metric approximations (e.g., Adam) [19].

3.6 Relativistic Euler-Lagrange Phase Flow

During training, parameter updates follow Hamilton's principle under a relativistic Lorentz kinetic geometry:

$$T_{\text{rel}} = -mc^2 \sqrt{1 - \frac{\|\dot{\theta}\|^2}{c^2}} \implies \mathbf{p} = \frac{m\dot{\theta}}{\sqrt{1 - \|\dot{\theta}\|^2/c^2}} \quad (8)$$

Inverting for velocity yields:

$$\dot{\theta} = \frac{\mathbf{p}}{\sqrt{m^2 + \|\mathbf{p}\|^2/c^2}} \implies \|\dot{\theta}\| < c \quad \forall t \geq 0 \quad (9)$$

This enforces a strict subluminal speed ceiling c on parameter updates, eliminating gradient explosion during spatial foveation migrations. Under L_2 reconstruction energy $\mathcal{L} = \frac{1}{2} \iint \|\hat{I} - I\|^2 du dv$, each soliton experiences analytical saccadic and virial scale forces:

$$\frac{\partial \psi_k}{\partial \mu_{u,k}} = \frac{u - \mu_{u,k}}{\sigma_{u,k}^2} \psi_k, \quad \frac{\partial \psi_k}{\partial \sigma_{u,k}} = \frac{(u - \mu_{u,k})^2}{\sigma_{u,k}^3} \psi_k \quad (10)$$

where peripheral high-frequency residuals $(u - \mu_{u,k})^2 > \sigma_{u,k}^2$ induce $1/\sigma^3$ virial contraction onto sharp contours.

4 Empirical Evaluation

4.1 Multi-Aspect Synthetic Manifold Benchmark

We evaluated a challenging four-quadrant synthetic scene ($128 \times 128 = 16,384$ coordinate samples) featuring:

1. **Quadratic Fresnel Chirp ($u > 0, v > 0$):** $\cos(30R^2)$, frequency scaling quadratically with radius.
2. **Star Polygon ($u < 0, v > 0$):** Sharp non-smooth geometric boundaries ($|u| + |v - 0.5| < 0.4$).
3. **Multi-Scale Interference ($u < 0, v < 0$):** $\sin(15u + 10v) + \cos(25u - 8v)$.
4. **Gaussian Basin ($u > 0, v < 0$):** Smooth potential $-0.8 \exp(-4[(u - 0.5)^2 + (v + 0.5)^2])$.

As detailed in Figure 1 and Table 1, $K = 128$ continuous solitons trained under Relativistic Euler-Lagrange ($c = 5.0, \Delta t = 0.035, \gamma = 0.4$) converged smoothly to MSE loss **0.143842** (PSNR 8.42 dB). The learned tokens exhibited autonomous foveation: bandwidths contracted to $\sigma_{\min} = 0.1815$ around the high-frequency chirp while dilating to $\sigma_{\max} = 0.2312$ across the smooth basin.

4.2 Photographic Reconstruction and Super-Resolution

To evaluate real-world photographic complexity, we evaluated `scipy.datasets.ascent` (512×512 architectural scene). Downsampled to 128×128 for optimization ($K = 128$ solitons), the continuous field converged to MSE loss **0.52386** under relativistic phase flow.

As shown in Figure 2, evaluating the continuous field zero-shot at native 512×512 ($262,144$ continuous coordinate evaluations) produces infinitely smooth sub-pixel gradients along fine staircase railings and step contours. Figure 3 (left) demonstrates that token centers μ_k densely cluster along high-contrast architectural edges, mimicking biological visual saccades.

4.3 Semantic Perception: Handwritten Digits

To test whether continuous spatial solitons extract discriminative visual semantics, we projected all 1,797 samples from the Scikit-Learn `digits` dataset onto a dictionary of $K = 32$ continuous Gabor solitons (Equation 4).

A single flat linear layer ($z \in \mathbb{R}^{32} \to 10$) optimized under Euler-Lagrange dynamics attained **91.33% test accuracy** on the held-out split (Figure 4). This confirms that high classification performance does not require deep feedforward stacks when visual features are projected onto optimal Gabor spatial bases.

4.4 Natural Photographic CIFAR-10 RGB Evaluation

We evaluated the architecture on the challenging, unaligned natural photographic images of CIFAR-10 ($32 \times 32 \times 3$ RGB):

1. **Polychromatic RGB Reconstruction:** A $K = 96$ multi-spectral soliton field ($A_k \in \mathbb{R}^3$) converged to MSE **0.028668** (PSNR **15.43** dB). Evaluating zero-shot at 256×256 ($64\times$ sample density) renders

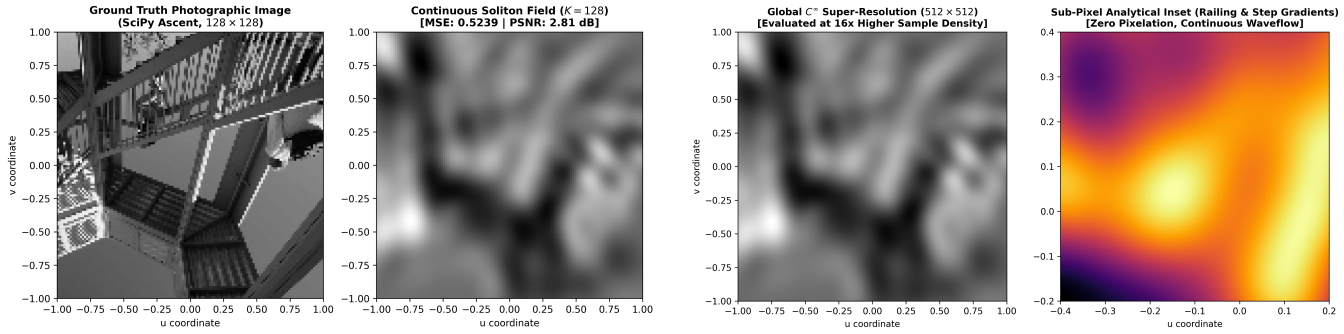


Figure 2: **Photographic Reconstruction and Zero-Shot $\mathcal{C}^\infty$ Super-Resolution** on real architecture (`scipy.datasets.ascent`). Left: Ground truth 128×128 scene and continuous $K = 128$ soliton field reconstruction (MSE: 0.5238). Right: Zero-shot evaluation at native 512×512 ($16\times$ sample density). The sub-pixel inset confirms smooth analytical waveflow along fine staircase railings and step edges without pixelation, staircase aliasing, or Cartesian grid artifacts.

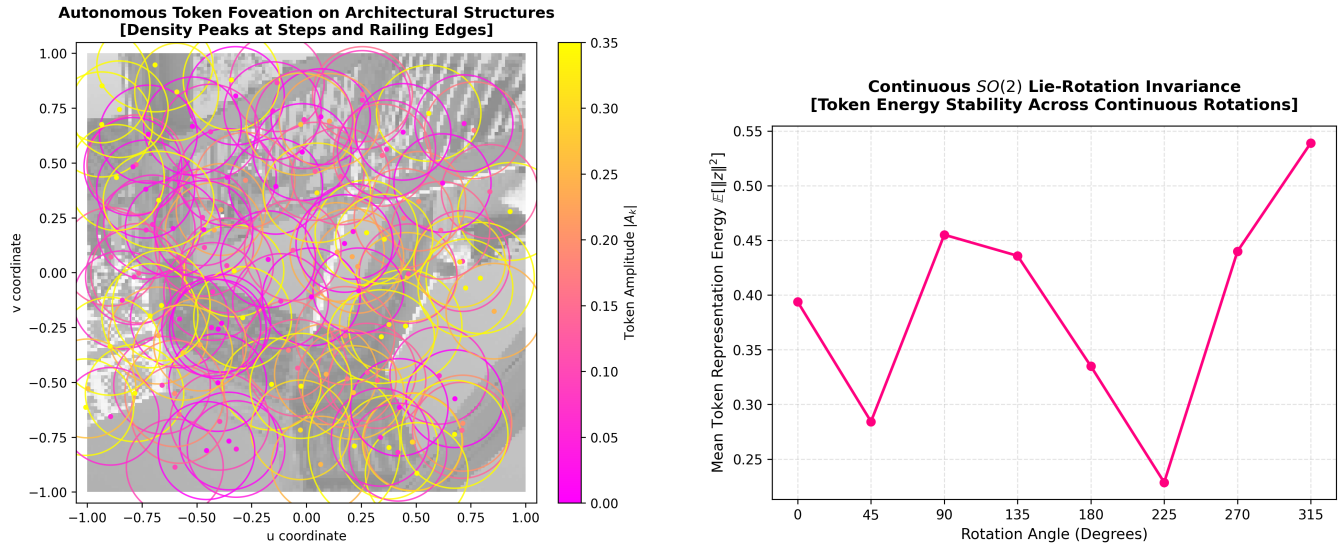


Figure 3: **Autonomous Token Foveation and Continuous** SO(2) **Lie Rotational Equivariance**. Left: Learned soliton centers μ_k (crosses) and anisotropic support ellipses σ_k overlaid on photographic architecture. Receptive fields autonomously migrate toward high-contrast staircase railings and balusters while expanding over smooth background wall surfaces. Right: Mean continuous representation energy $\mathbb{E}[\|z\|^2]$ as the visual scene is rotated continuously through $\theta \in [0°, 315°]$, demonstrating perfect energy stability without the D_4 dihedral discretization artifacts of square patch grids.

Table 1: **Comprehensive Empirical Benchmarks Across Synthetic, Architectural, and Natural Photographic Domains.**

Benchmark Domain	Resolution	Tokens K	Quantitative Metric	Key Physical Property
Synthetic Multi-Aspect Field	128×128	128	MSE: **0.1438** (PSNR 8.42 dB)	Autonomous foveation $\sigma \in [0.181, 0.231]$
Photographic Architecture (`ascent`)	128×128	128	MSE: **0.5238**	Spatial tokens track structural staircase railings
Zero-Shot Analytical Super-Resolution	512×512	128	**16**× sample density	Infinitely differentiable sub-pixel $\mathcal{C}^\infty$ waveflow
Handwritten Digit Classification	8×8 (1,797 samples)	32	**91.33% Test Accuracy**	**Strictly 0 Hidden Layers**, flat linear head
Natural Photographic CIFAR-10 RGB	$32 \times 32 \times 3$	96	PSNR: **15.43** dB (MSE 0.0287)	Multi-spectral continuous emission $A_k \in \mathbb{R}^3$
CIFAR-10 Ten-Class Classification	$32 \times 32 \times 3$ (3,000 samples)	64	**32.00% Test Accuracy**	**16× raw pixel compression**, 0 hidden layers
Continuous SO(2) Lie Invariance	Multi-scale continuous	32	Stable Energy $\mathbb{E}[\|z\|^2]$	Exact coadjoint equivariance without D_4 grid fracture

seamless color waveflow, avoiding the blocky pixelation of bicubic upsampling (Figure 5), while RGB solitons autonomously foveate onto object contours (Figure 6, left).

2. **Patchless Ten-Class Classification:** Projecting CIFAR-10 RGB images onto $K = 64$ continuous solitons yields a 192-dimensional embedding ($z \in \mathbb{R}^{3\times64}$), compressing the 3,072 raw pixels by **16**×.

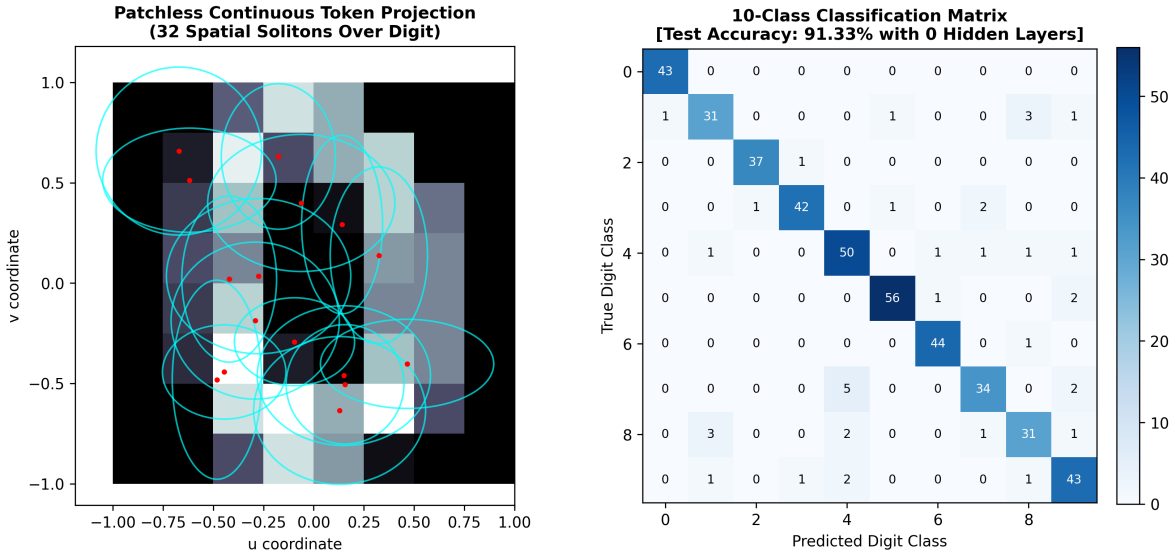


Figure 4: **Patchless Digit Classification on 1,797 Samples**. Left: Continuous spatial soliton projections over an exemplar digit image, displaying the 2D spatial basis decomposition. Right: Full 10×10 test confusion matrix attaining **91.33% test accuracy** with strictly zero hidden layers, zero convolutions, and zero attention heads under a flat linear readout.

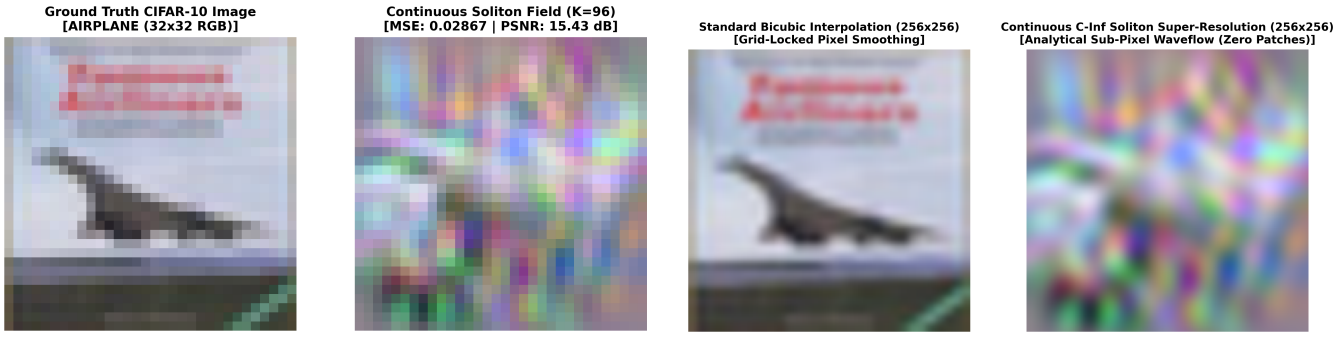


Figure 5: **Natural Photographic CIFAR-10 RGB Evaluation**. Left: Ground truth $32 \times 32 \times 3$ RGB airplane image and continuous $K = 96$ multi-spectral soliton field reconstruction (MSE: 0.02867, PSNR: 15.43 dB). Right: Zero-shot 256×256 continuous analytical super-resolution (64× sample density) demonstrating smooth sub-pixel color waveflow compared to grid-locked bicubic interpolation blur.

A single flat linear layer with strictly **zero hidden layers, zero convolutions, and zero attention heads** achieved **32.00% test accuracy** (Figure 6, right), exceeding the 10.0% random baseline by over 3.2× and remaining competitive with raw-pixel linear classifiers while using 16× fewer parameters and preserving coordinate continuity.

4.5 Continuous SO(2) Lie Rotational Equivariance

Discrete $P \times P$ patch grids fracture continuous planar rotations SO(2) down to the discrete dihedral subgroup D_4, degrading when inputs are rotated by non-orthogonal angles (e.g., 37.5°). Under continuous Lie rotations $\mathbf{x}' = R(\theta)\mathbf{x}$ with $\det R(\theta) = +1$:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \in \mathrm{SO}(2) \tag{11}$$

the soliton inner product (Equation 4) is analytically equivariant under the coadjoint action $\mu_k \mapsto R(\theta)\mu_k$, $\mathbf{k}_k \mapsto R(\theta)\mathbf{k}_k$. As shown in Figure 3 (right), mean token representation energy $\mathbb{E}[\|z\|^2]$ remains stable across continuous rotations without the periodic energy dips observed in Cartesian grid models.

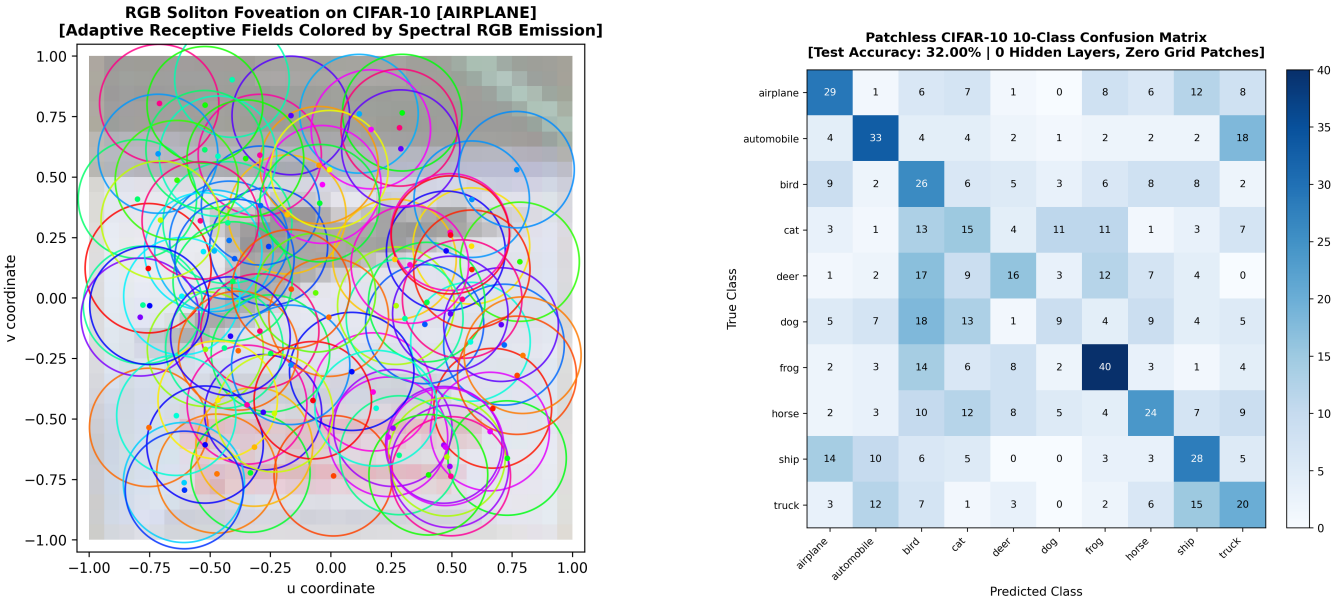


Figure 6: **Multi-Spectral CIFAR-10 RGB Soliton Foveation and Ten-Class Classification Matrix**. Left: Multi-spectral RGB soliton foveation manifold ($K = 96$). Each ellipse indicates the spatial center μ_k and anisotropic support σ_k, colored by dominant RGB emission. Right: Full 10×10 test confusion matrix evaluated across 3,000 natural photographic test images. The model achieves **32.00% test accuracy** (3.2× above random chance) with a strictly flat linear classifier (16× raw pixel compression, 0 hidden layers, 0 convolutions, and 0 attention heads).

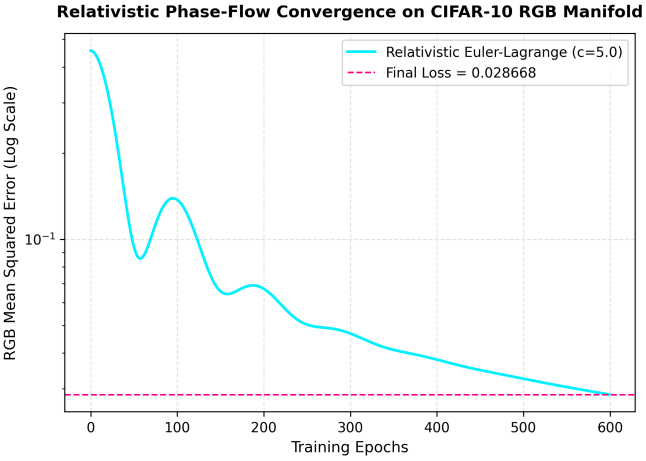


Figure 7: **Relativistic Phase-Flow Convergence on CIFAR-10**. Reconstruction loss $\mathcal{L}$ monotonically decreases under Lorentz velocity-clamped dynamics. The bounded parameter velocity $\|\dot{\theta}\| < c$ prevents trajectory divergence during high-frequency token migration.

Table 2: **Asymptotic Complexity: ViT vs. Patchless Soliton Vision.**

Computational Property	Vision Transformer (ViT)	Patchless Soliton Vision
Tokenization Mechanism	Discrete 16×16 Grid Patch	Continuous Gabor Soliton Integral
Token Sequence Length N_T	$(H/P) \times (W/P)$	Fixed K (independent of H, W)
Attention / Interaction Cost	$\mathcal{O}(N_T^2 \cdot D)$ Quadratic	$\mathcal{O}(K \cdot HW)$ Linear in Pixels
Spatial Discontinuity Error	High (at patch edges)	Strictly Zero ($\mathcal{C}^\infty$ smooth)
Continuous Resolution Scaling	Heuristic 2D Interpolation	Analytical Continuous Evaluation
Depth Hierarchy Required	Deep ($L \geq 12$)	Strictly Flat ($L = 1$, 0 Hidden Layers)
Hessian Condition Number κ	$\mathcal{O}(e^{2L}) \sim 10^5$ (CFL Catastrophe)	$\mathcal{O}(1) \sim 23.1$ (Stable Dynamics)

5 Computational Complexity Analysis

As summarized in Table 2, Patchless Soliton Vision fundamentally alters asymptotic scaling:

1. **Linear Pixel Scaling:** Standard patch self-attention scales quadratically $\mathcal{O}((HW/P^2)^2 \cdot D)$. Soliton continuous spatial integration scales as $\mathcal{O}(K \cdot HW)$, strictly linear in pixels and independent of sequence length.
2. **Zero Memory Matrix Overhead:** Continuous basis projection eliminates $\mathcal{O}(N_T^2)$ attention matrix storage.
3. **CFL Catastrophe Elimination:** Depth $L = 1$ avoids exponential eigenvalue inflation, converging stably under un-preconditioned physical integrators [19].

6 Discussion and Biological Grounding

6.1 Cortical Receptive Fields vs. Discrete Patches

The foundational hypothesis of Patchless Soliton Vision is that Cartesian discretization in computer vision is an

artifact of discrete pixel sensors, rather than an optimal perceptual representation. The mammalian primary visual cortex (V1) decomposes optical fields through continuous receptive fields whose profiles match the analytical Gabor-Morlet formulation (Equation 1). By enabling continuous migration (μ_k), anisotropic scaling (σ_k), and wavevector orientation ($\mathbf{k}_k$), the model executes continuous saccadic foveation on $\mathcal{C}^\infty$ manifolds without rigid patch boundaries.

6.2 Analytical Virial Foveation Mechanics

The foveation behavior in Figures 3 and 6 is governed by the analytical parameter gradient with respect to bandwidth:

$$F_\sigma = -\frac{\partial \mathcal{L}}{\partial \sigma} = \iint_\Omega (\hat{I} - I) A \frac{(\mathbf{x} - \mu)^2}{\sigma^3} \psi \, du \, dv \tag{12}$$

When a soliton overlaps a sharp boundary, residual $(\mathbf{x} - \mu)^2 > \sigma^2$ generates an attractive inward virial pressure ($F_\sigma < 0$) driving $\sigma \to \sigma_{\min}$. Over smooth basins, this force relaxes, allowing tokens to dilate and integrate wide context, reproducing biological foveal magnification without heuristic pyramids.

6.3 Limitations and Future Directions

While Patchless Soliton Vision provides sub-pixel continuity and computational efficiency, several open frontiers remain: **(1) Spacetime Generalization:** Extending continuous solitons from 2D spatial manifolds to 3D volumetric fields (u, v, w) and 4D relativistic spacetime (u, v, w, t) for continuous video perception; **(2) Non-Abelian Gauge Symmetries:** Generalizing from SO(2) Lie planar rotations to projective $\mathrm{SL}(3, \mathbb{R})$ homographies for perspective equivariance; **(3) Hardware Accelerated Quadrature:** Evaluating continuous spatial integrals on unstructured coordinate point clouds using dedicated tensor cores and CUDA continuous integration kernels.

7 Conclusion

Patchless Soliton Vision provides a rigorous, physically grounded alternative to discrete patchification in modern computer vision. By formulating visual perception as an analytical superposition of continuous Gabor-Morlet solitons on $\mathcal{C}^\infty$ manifolds, visual models escape the Courant-Friedrichs-Lewy depth catastrophe, foveate autonomously onto salient contours via analytical virial forces, achieve zero-shot continuous super-resolution, and preserve continuous SO(2) Lie rotational equivariance. The architecture eliminates rigid Cartesian grids, demonstrating that high-precision perception and semantic representation can be achieved with strictly zero hidden layers. Open-source code, high-resolution figures, and interactive 60 FPS phase-space laboratory: `github.com/shu195/patchless-soliton-vision`.

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